Price Elasticity of Demand (PED)

Exam code: 9BS0

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An introduction to price elasticity of demand

- When there is an **increase in price**, there will be a **fall in the quantity demanded (QD)**, and when there is a **fall in price**, there will be an **increase in the QD**
  - The question businesses are interested in is "**by how much** will **the QD change?**"
- **Price elasticity of demand** (PED) helps us to calculate **how responsive the change in QD will be to a change in price**
  - **The responsiveness is different for different types of products**

Calculation of PED

- **The PED value is always negative**
- **PED can be calculated** using the following formula:

$$\text{PED} = \frac{\text{\% change in quantity demanded}}{\text{\% change in price}} = \frac{\text{\% } \Delta \text{ in QD}}{\text{\% } \Delta \text{ in P}}$$

- **To calculate a % change**, use the following formula:

$$\text{\% Change} = \frac{\text{new value} - \text{old value}}{\text{old value}} \times 100$$

Worked Example

The PED for popcorn at the cinema is –0.8. The current price of a box of popcorn is £5. Using the data, calculate the percentage change in QD following a £1 increase in the price of a box of popcorn. You are advised to show your work.

[4]

Step 1: State the PED formula:

$$\text{PED} = \frac{\text{\% change in demand}}{\text{\% change in price}}$$ [1]

Step 2: Calculate the percentage change in price:

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

$$= 20\%$$

Step 3: Insert the data you have been given into the formula:

$$-0.8 = \frac{x}{20\%}$$ [1]

Step 4: Rearrange and solve for x:

$$x = -0.8 \times 20$$ [1]

$$x = -16\%$$

Help me understand

Examiner Tips and Tricks

Remember, **if the price decreases, QD increases**, and **if the price increases, QD decreases**. In this case, price increases. Therefore, QD must fall.

Interpretation of PED numerical values

- The numerical value of PED indicates the **responsiveness** of a change in QD to a change in price
- PED will always be negative due to the **inverse relationship** between price and quantity
  - If the price goes up, the QD goes down
  - If the price goes down, the QD goes up

Interpretation of PED values

Numerical value	Type of good	Explanation
-1 or lower	**Elastic** • Examples include **luxury products** such as cars, smart watches, foreign holidays, cinema visits, jewellery and branded goods	• Demand is **more responsive to a change in price** • The %Δ in QD is **more than proportional** to the %Δ in P

	necessities such as bread, milk, eggs, potatoes, fuel, rent and toothpaste • **Addictive products** such as cigarettes and sugary foods	• The %Δ in QD is **less than proportional** to the %Δ in P

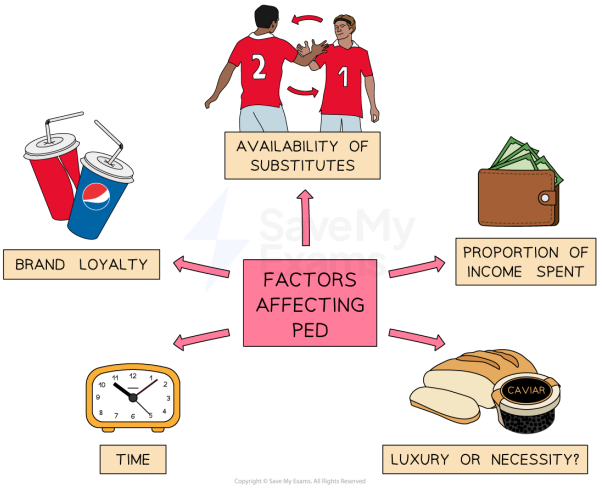
Examiner Tips and Tricks

Students often confuse the negative sign in the answer to PED questions. Do not assume that the negative is mathematical, such that an elasticity of −1 is smaller than −0.3. It is larger (more price elastic).

The **PED will always be negative, indicating the inverse relationship between price and QD** (i.e. when the price increases, QD decreases, and when the price decreases, QD increases).

When interpreting the value of PED, do not say that "the product is elastic or inelastic"; it is better to say that "**demand** for the product is price elastic or price inelastic".

Factors influencing the PED



The factors that determine whether the demand for a product is more price elastic or price inelastic

Brand loyalty

- The aim of advertising and marketing expenditure by a business is to shift the demand curve to the right and **make the demand more price inelastic**

Availability of substitutes

- PED will be more price inelastic (lower) for goods that have **fewer substitutes**
  - For example, petrol has fewer substitutes and is more price inelastic, whereas chocolate bars have more substitutes and are more price elastic

The proportion of income taken up by the product

- The **smaller the proportion of income** we spend on a product, the **more price inelastic** the demand will be
  - For example, a small amount of income is spent on salt, so the demand for salt will be more price inelastic. However, buying a new car takes up a **bigger proportion** of consumer income, so **the PED is higher**

Luxury or necessity

- **Necessities** are required as part of consumers' daily needs, and therefore, demand for them is **more price inelastic**
  - For example, bread, milk, petrol, gas and electricity might be considered necessities
- **Luxuries** are not essential, and therefore, demand for them is **more price elastic**
  - For example, smoked salmon, Nike Air Jordans and foreign holidays might be considered luxuries

Time

- The **longer the time period** under consideration, the more price elastic the demand for a good or service is likely to be (consumers have more time to search for substitutes)
- The **shorter the time period** under consideration, the more price inelastic the demand for a good or service is likely to be
  - For example, if the price of petrol increases, making driving more expensive, there is little that consumers can do in the short term. However, they may switch to alternatives such as public transport or bicycles in the long term

The significance of PED to businesses

- If businesses can **determine the PED** for their products, they can adjust their **pricing strategy** to **maximise their revenue**

The relationship between PED and total revenue

(PED is less than or equal to 1), raising the price will lead to an **increase in total revenue**. However, lowering the price will lead to a **fall in total revenue**

Price elastic demand

- PED is **greater than 1**
- An increase in the selling price reduces the total amount of revenue generated from sales
- A reduction in the selling price increases the total amount of revenue generated from sales
- **Competitive pricing strategies** are best employed for products that are price elastic in demand

Price inelastic demand

- If demand for their products is relatively **price elastic (PED is greater than or less than 1)**, raising the price will lead to a **fall in total revenue**
- However, lowering the price will lead to a **rise in total revenue**

Price inelastic demand

- PED is **between 0 and 1**
- An increase in the selling price increases the total amount of revenue generated from sales
- A reduction in the selling price reduces the total amount of revenue generated from sales
- **Price skimming strategies** are best employed for products that are price inelastic in demand

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